

85

Me

**three fillers,
multiple
indications**

the element for skincare experts

densiMatrix® technology:

mesofiller® is the range of dermal fillers based on cross-linked hyaluronic acid of non-animal origin obtained by biofermentation, developed using a unique technology which ensures maximum quality, safety* and efficacy in the clinical practice.

Highly-effective cross-linking and purification

High degree of modification

Several HA concentrations with homogeneous cross-linking, without free HA.

Strict standardisation⁽¹⁾

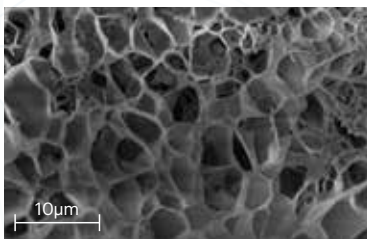
Control of key parameters in order to ensure product quality.

Maximum safety⁽²⁾

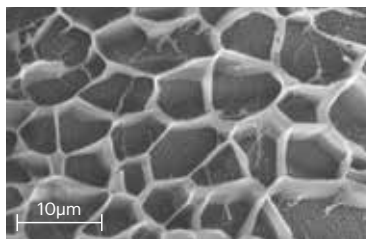
Unique purification process that ensures minimal residual value of BDDE.⁽³⁾

The result is a structure without breakages, evidencing high-quality cross-linking and homogeneity:

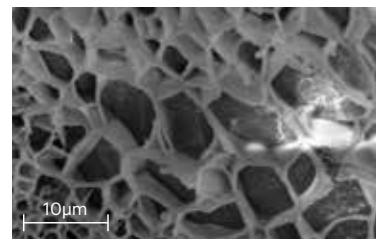
mesofiller® periocular



mesofiller® global



mesofiller® intense



Images obtained by electron microscopy (X2000) ⁽⁴⁾.

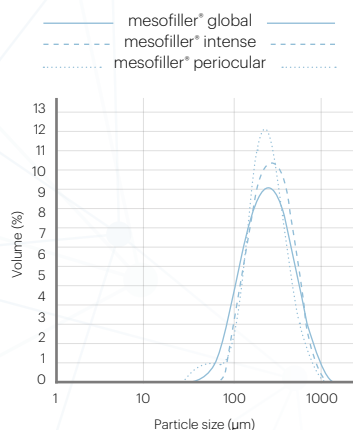
Particle size and homogeneity

Homogeneous particles

Particle size distribution with very small variability.

Particles smaller than needle diameter:

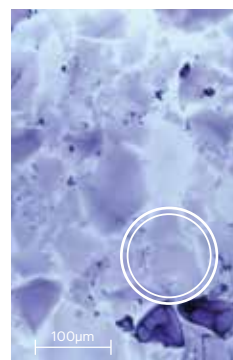
Mean particle size smaller than 400 microns (needle diameter 27G), that prevents deformation during extrusion.⁽⁶⁾



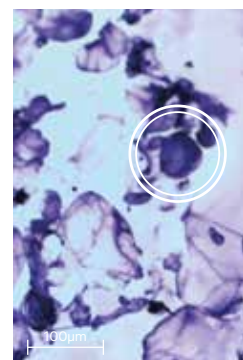
Images obtained by Laser Diffraction (LD) ⁽⁵⁾.



mesofiller®
periocular



mesofiller®
global



mesofiller®
intense

Images obtained by optical microscopy with methylene blue stain (X4) ⁽⁷⁾.

*Manufactured following GMP standards and in compliance with ISO 13485 standard. Therefore, mesofiller bears CE 0373 mark, in compliance with the current European regulations on medical devices.

Durability, safety and improvement of tissue quality

High durability

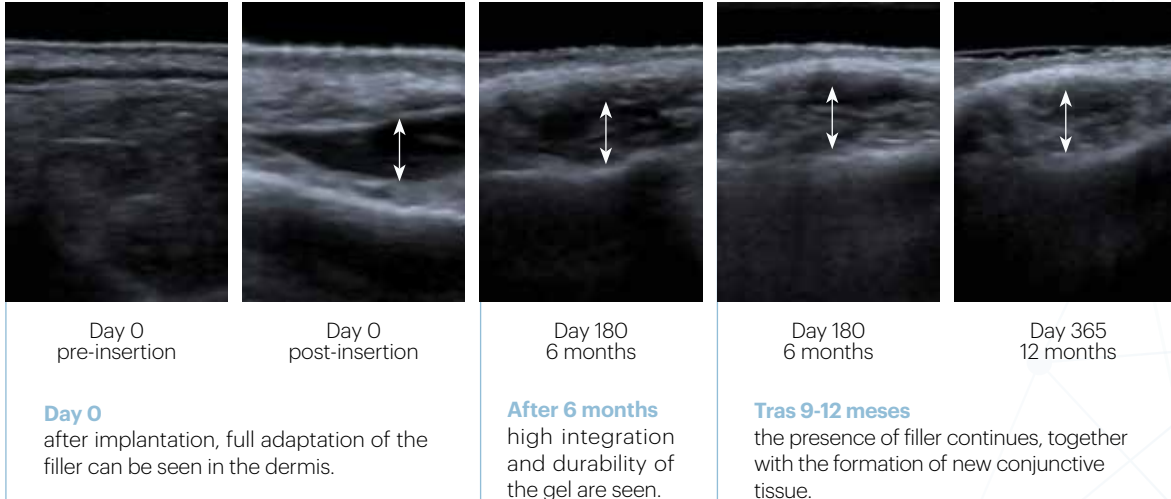
High resistance to enzyme hydrolysis with permanence in the tissue for up to 12 months.

Tissue repair⁽⁸⁾

Full integration that stimulates regeneration pathways, improving tissue quality.

Absence of nodules

The high bioaffinity and quality of the gel prevents nodular reactions.



Over 12 months no presence of nodules is seen (anechoic areas).

Images obtained using ultrasound equipment (Toshiba Xario 200 18 MHz multifrequency linear probe).⁽⁹⁾

Characteristics of mesofiller® in the clinical practice:



High purity in a single stage



High uniformity of the gel:
lower risk of nodules



Completely hydrated
*very low swelling factor***



Preservation of viscoelastic properties after tissue implantation



Maximum safety:
among the lowest BDDE values on the market



Durability from 9 to 12 months



Low, constant necessary extrusion force



Stimulation of tissue repair mechanisms

**degree of gel swelling: capacity to absorb water under physiological conditions. It is related to the degree of hydration present in the gel preparation.⁽¹⁰⁾

dermal filler medical solutions

New mesofiller prefilled syringe

Detailed 0.05 ml graduation

Improved accuracy in the injection

Improved stability

Maximum stability and control during the injection

Monomaterial syringe

Further compatibility and resistance to extrusion force

luer-lock connection



contains:

1 prefilled syringe x 1 ml

2 sterile syringes 30G ½"

Recommended cannula: 25G 2"

mesofiller® periocular

DERMAL FILLER SOLUTIONS

15 mg/ml

Reticulated hyaluronic acid gel specific for treating imperfections in the periorbital area.

- Reduces imperfections in the periocular area caused by ageing, fatigue and high gesticulation.
- Ideal rheological profile suitable for the area, providing further smoothness and perfect adaptation to the tissue.
- Low swelling factor for a more accurate and predictable result. Minimises the formation of oedemas.

Application areas:

External periorbital wrinkles
Nasojugal line and palpebromalar groove
Eyebrow tail

Suitable for:

Forehead
Lip moisturising



contains:

1 prefilled syringe x 1 ml
2 sterile syringes 27G ½"

mesofiller® global

DERMAL FILLER SOLUTIONS

20 mg/ml

Reticulated hyaluronic acid gel that provides a perfect balance between elasticity and viscosity to fill and correct multiple indications.

- Its rheological profile offers a highly versatile soft and elastic filler.
- Its smoothness helps restore volumes with a natural result, perfectly adapting to the shapes of the face without affecting the habitual facial expressions.

Application areas:

Medium wrinkles
Lip augmentation and profiling
Around the mouth
Marionette lines
Moderate nasolabial fold
Temporal area
Glabella
Cheeks
Earlobes
Nose modelling: ridge, supratip

Suitable for:

Hands
Hypotrophic scars



contains:

1 prefilled syringe x 1 ml
2 sterile syringes 27G ½"

mesofiller® intense

DERMAL FILLER SOLUTIONS

25 mg/ml

High-concentration reticulated hyaluronic acid gel, especially recommended to reshape facial volumes.

- Its rheological profile is noted for being a more viscous and rigid gel, capable of remaining in specific areas.
- It provides high supporting capacity for optimal projection and elevation of the tissue.

Application areas:

Deep wrinkles
Facioplasty -face lift-
Temporal area
Malar area
Severe nasolabial fold
Jawline

Cheeks
Chin
Nose modelling: nasal tip, columella, columellar-labial angle, nasion

dermal filler medical solutions

mesofiller®: three rheological profiles to cover multiple indications

Innovation in production technology, together with the full balance of viscoelastic properties, ensures obtaining three gels with differentiated characteristics for maximum versatility in application.



mesofiller® periocular



mesofiller® global

Concentration [HA]	15 mg/ml	20 mg/ml
G' Elastic module	115 Pa	210 Pa
G'' Viscous module	17 Pa	30 Pa
Swelling factor	1,9 - 2,4 ml/g (very low)***	1,9 - 2,4 ml/g (very low)***

Clinical implications⁽¹⁾

High G''

- Lower spreadability, greater tendency to remain localized.
- More precise remodeling capacity.

+

viscous module (G'')

Low G''

- Higher spreadability, greater adaptability to tissue ("spread capacity").
- Higher softness, a more ductile filler.

-

Low G'

- Lower capacity to tissue lifting.
- More suitable for areas which require less resistance to tissue.
- Higher softness, less palpable.

mesofiller® periocular
[AH] 15 mg/ml

mesofiller® global
[AH] 20 mg/ml

elastic module (G')

***measured under the same conditions and compared to other market values: Steven Fagien, M.D. Vince Bertucci, M.D. Rheologic and Physicochemical Properties Used to Differentiate Injectable Hyaluronic Acid Filler Products. Plastic and Reconstructive Surgery, HYPERLINK 143 (4), 707e-720e, 2019.

Application areas:

The three products of the mesofiller® range adapt to the different face areas to correct, highlight and define, achieving a long-lasting, natural outcome.



mesofiller® intense

25 mg/ml

320 Pa

44 Pa

1,9 - 2,4 ml/g

(very low)***



mesofiller® intense
[AH] 25 mg/ml



High G'

- Higher capacity to tissue lifting and projection ("lifting capacity").
- Higher resistance to movements of facial musculature and tissue.
- Higher stiffness, more palpable.

- P** frontal area
- G** glabella
- I** eyebrows
- I G** temporal fossa
- P** periocular area
- I G** nose
- G** earlobe
- I** malar area
- I G** cheeks
- I G** nasolabial fold
- G** perioral area and commissure
- G** lip
- I** facial oval
- I** chin

mesofiller® is also suitable for dermal filling in body areas such as décolleté, hands or hypertrophic scars.

P mesofiller® periocular **G** mesofiller® global **I** mesofiller® intense

mesofiller®: clinical cases

Treatment of periocular bags and wrinkles

mesofiller® periocular



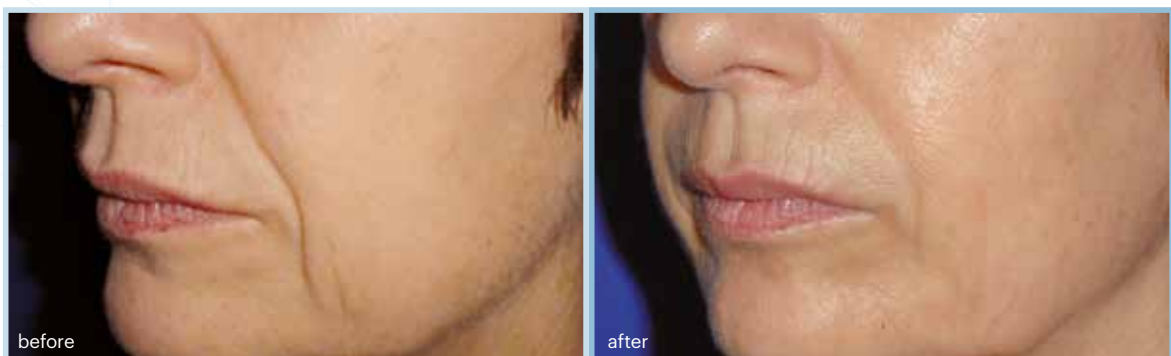
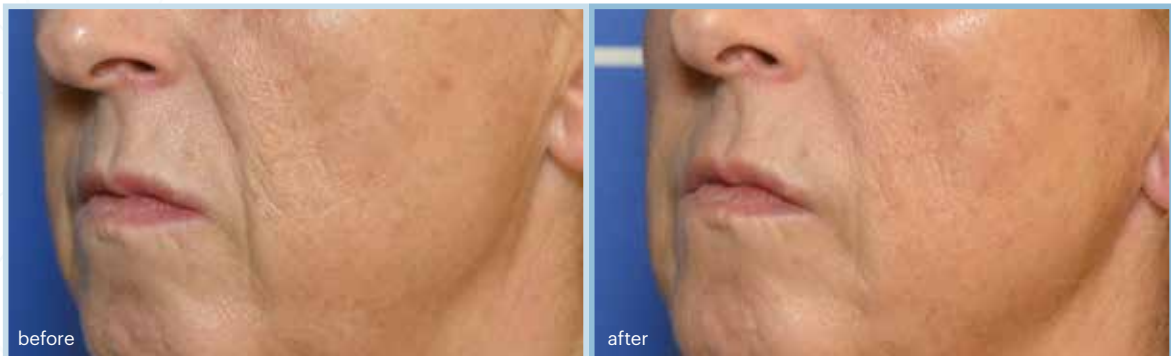
Treatments of under-eye bags, lower eyelid drooping and periocular wrinkles

mesofiller® periocular



Correction of the nasolabial fold

mesofiller® global y mesofiller® intense



mesofiller®: clinical cases

Lip volume increase

mesofiller® global



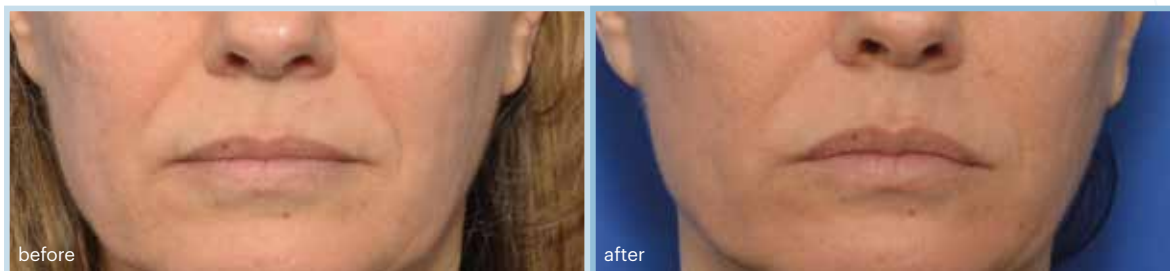
Volume increase in lips and commissure****

mesofiller® global



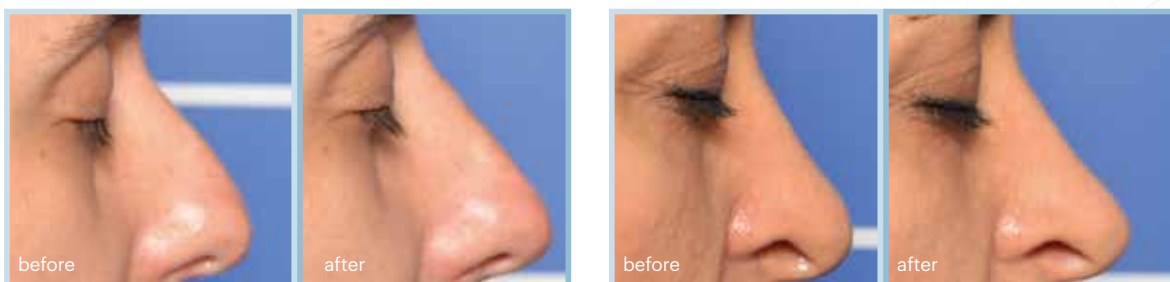
Filler in cheeks, treatment of wrinkles in nasolabial fold and perioral area

mesofiller® global y mesofiller® intense



Nose shaping

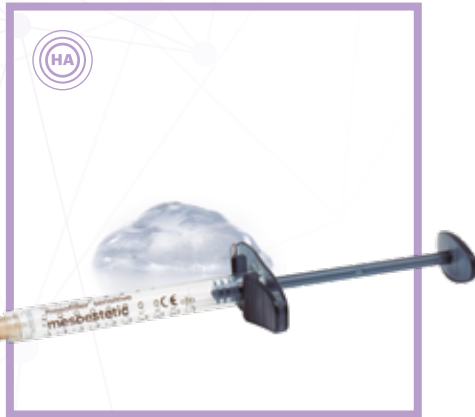
mesofiller® global y mesofiller® intense



****Images courtesy of Dr. Gabriel Ayala.

▽ intimate filler medical solutions

Products with specific rheological profiles to moisturise and restore women's intimate area adapting to the viscoelastic properties of each application area.



1 syringe X 1 ml

mesofiller® sensitive

INTIMATE FILLER MEDICAL SOLUTIONS

Cross-linked hyaluronic acid gel for the treatment of vaginal atrophy, dryness, dyspareunia and sexual dysfunction.

Concentration [cross-linked HA]

20 mg/ml

G' Elastic module

196 Pa

G'' Viscous module

25 Pa

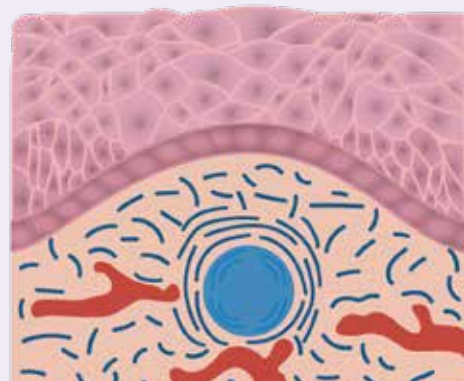
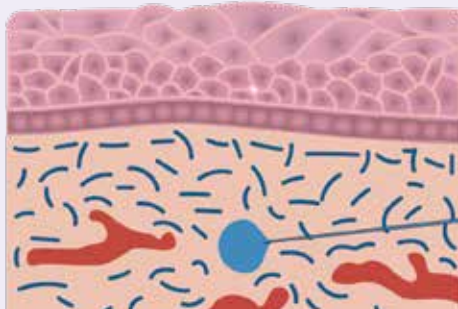
Application areas

vaginal vestibule, introitus,
vaginal walls, clitoris, G spot

How does it work?

It improves functionality of the vaginal mucosa, promotes regeneration of atrophic tissue and increases the sensitivity of erogenous zones.

Vaginal mucosa



Improves hydration of the vaginal mucosa. Enhances transport and cell trophic function. Stimulates angiogenesis. Activates collagen and elastin synthesis. Increases epithelial thickness. Restores the volume of erogenous zones enhancing friction during SR (clitoris and G spot).⁽³⁾



2 syringes X 1 ml

mesofiller® sensitive plus

INTIMATE FILLER MEDICAL SOLUTIONS

Cross-linked hyaluronic acid gel for the treatment and correction of vulvar ptosis and hypertrophy in the external genital area.

Concentration [cross-linked HA]

25 mg/ml

G' Elastic module

305 Pa

G'' Viscous module

40 Pa

Application areas

labia majora

How does it work?

It fills and restores tissue volume in the labia majora to preserve their protective function and improve their appearance.



single-dose 12 x 5 ml

hydragel intimate

MOISTURISING SOLUTIONS

Moisturising gel for women's intimate area.

Moisturiser

Hyaluronic acid at different molecular weights to return hydration and juiciness to the tissue.

Rejuvenating

It contains encapsulated retinol that activates cell differentiation.

Biorebalancer

It helps maintain the natural balance of the area due to the presence of a biopolysaccharide.



DERMAL FILLER MEDICAL SOLUTIONS

mesofiller® provides
**three concentrations
of cross-linked HA** for
approaching **multiple
indications**.

Unique technology
densimatrix leads to
**optimum rheological
characteristics** for the
clinical practice.

Maximum safety with
**minimum residual
BDDE levels.**

Excellent durability demonstrated using ultrasound tests.



1. Biao Yanga, Xueping Guoa; Determination of modification degree in BDDE-modified hyaluronic acid hydrogel by SEC/MS. 2015 **2.** Fidalgo J, Deglesne PA, Arroyo R, Sepúlveda L, Ranvea E, Deprez P. Detection of a new reaction by-product in BDDE cross-linked autoclaved hyaluronic acid hydrogels by LC-MS analysis. Med Devices (Auckl). 2018;11:367-376. **3.** Departamento de calidad mesoestetic Pharma Group.s.l análisis por cromatografía de gases (HPLC) 2019 **4.** Departamento de I+D+I mesoestetic Pharma Group.s.l análisis microscopia electrónica mesofiller® 2020 **5.** Departamento de I+D+I mesoestetic Pharma Group.s.l Distribución del tamaño de partícula mesofiller® 2020 **6.** Won Lee, MD, PhD, WookOh, MD, 2019 Soft Tissue Filler Properties Can Be Altered by a Small-Diameter Needle **7.** Departamento de I+D+I mesoestetic Pharma Group.s.l análisis microscopia óptica mesofiller® 2020 **8.** La Gatta A, Salzillo. RHyaluronan-based hydrogels as dermal fillers: The biophysical properties that translate into a "volumetric" effect. 2019;14(6):e0218287. **9.** Departamento medico mesoestetic Pharma Group.s.l, Estudio ecográfico fillers 2020 **10.** Steven Fagien, M.D. Vince Bertucci, M.D. Rheologic and Physicochemical Properties Used to Differentiate Injectable Hyaluronic Acid Filler Products. 2019 **11.** Sundaram H, Cassuto D. Biophysical characteristics of hyaluronic acid soft-tissue fillers and their relevance to aesthetic applications. 2013 Oct;132(4 Suppl 2):S5-21S.



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KF-MILO088/2205